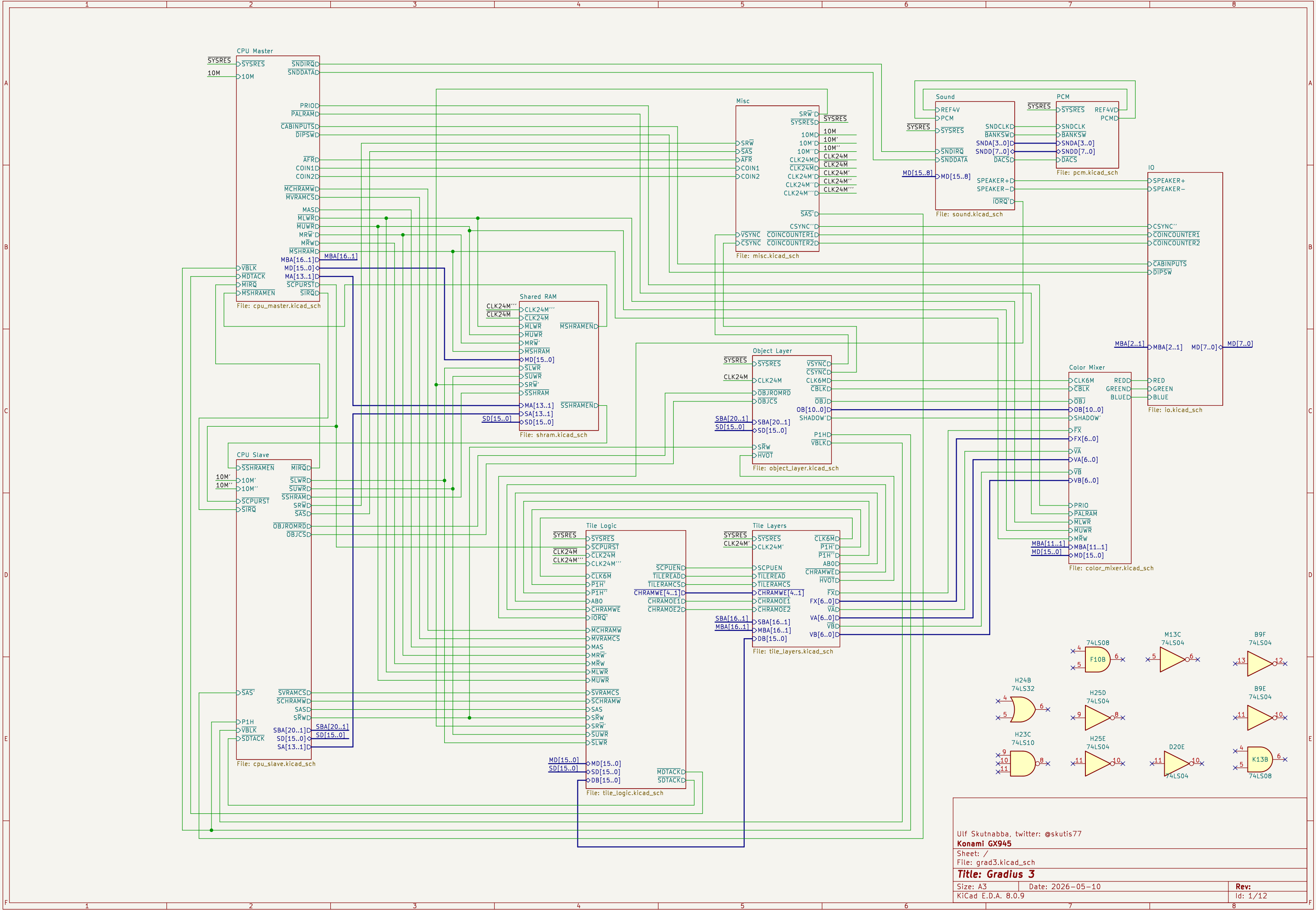
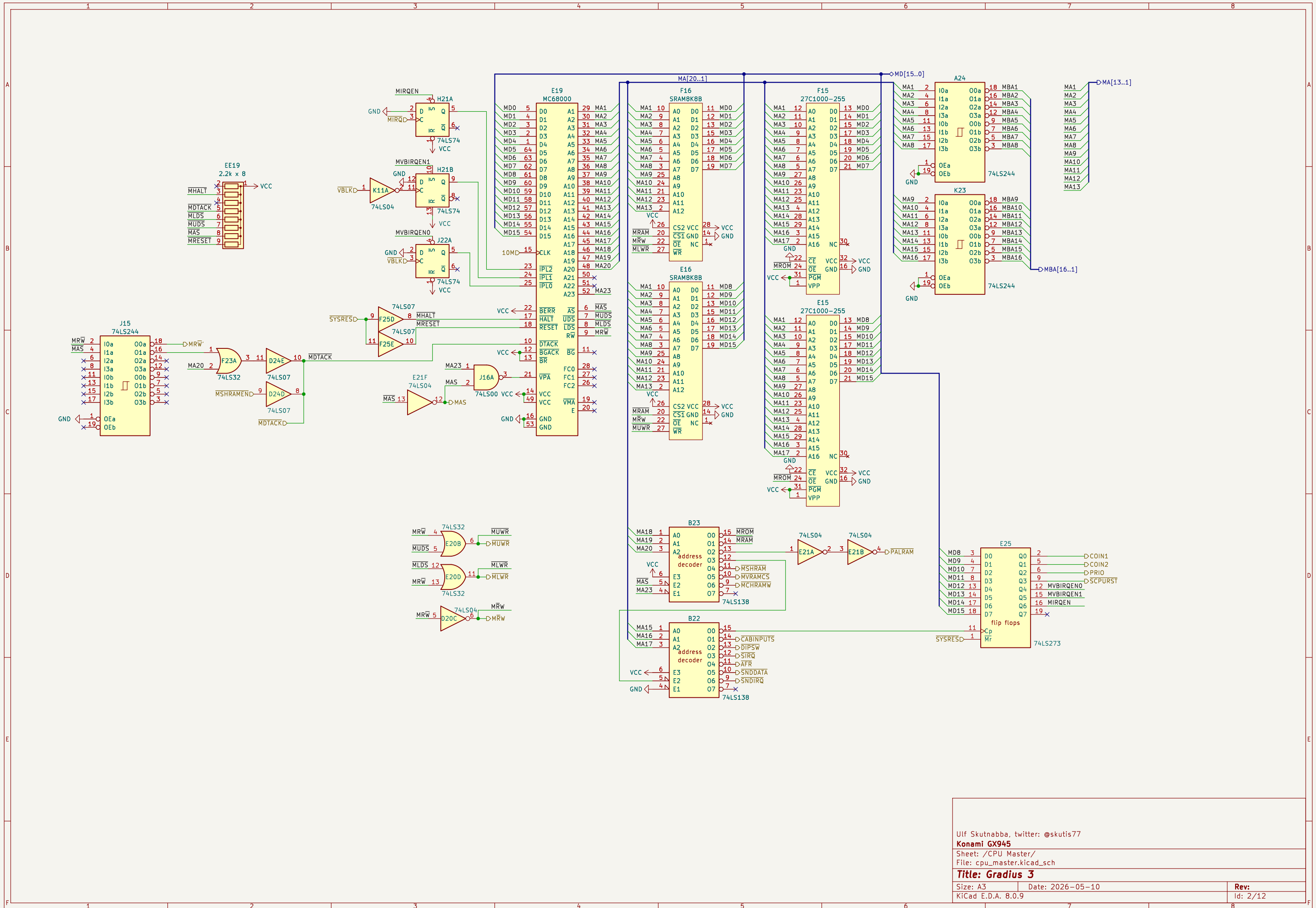
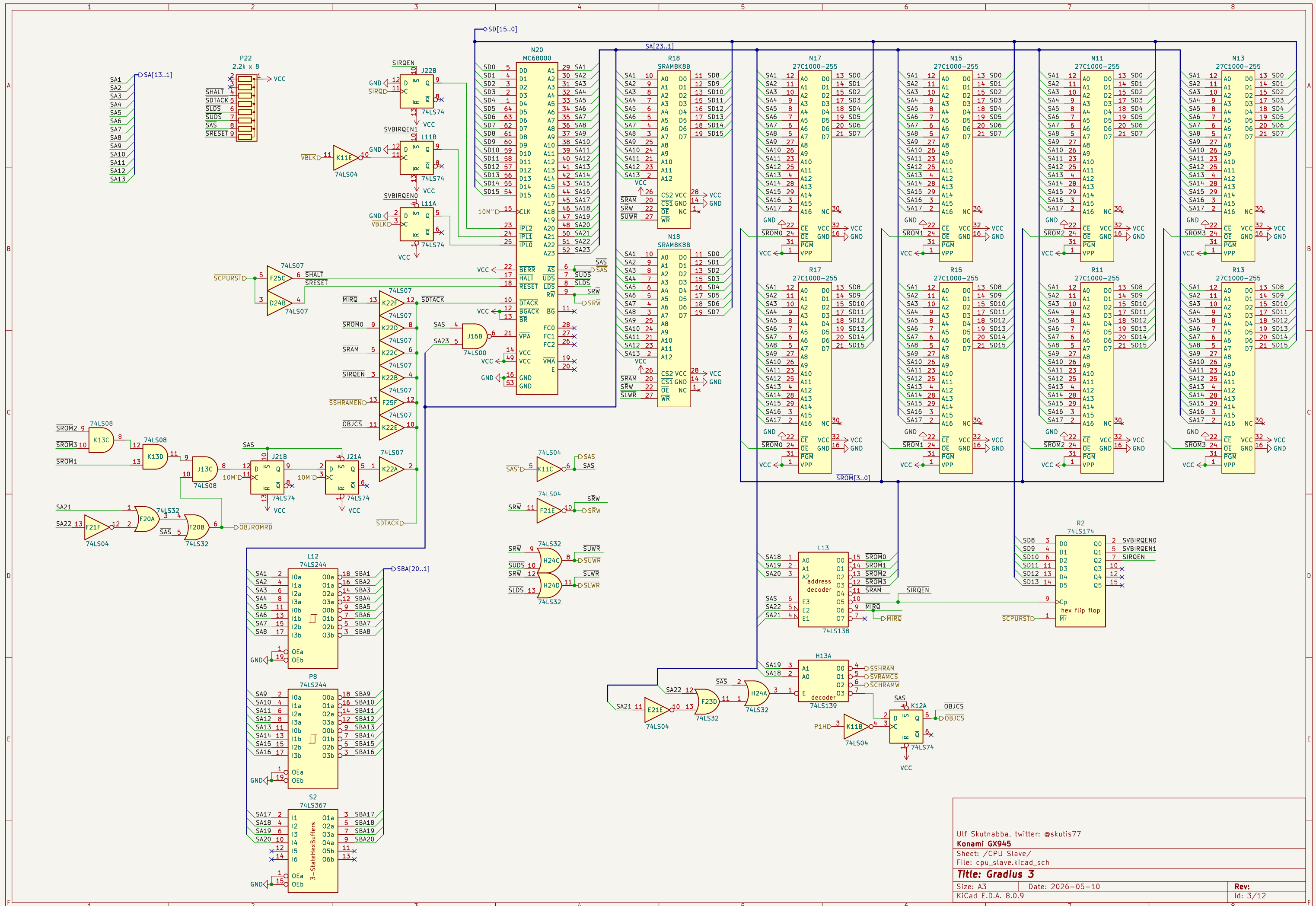
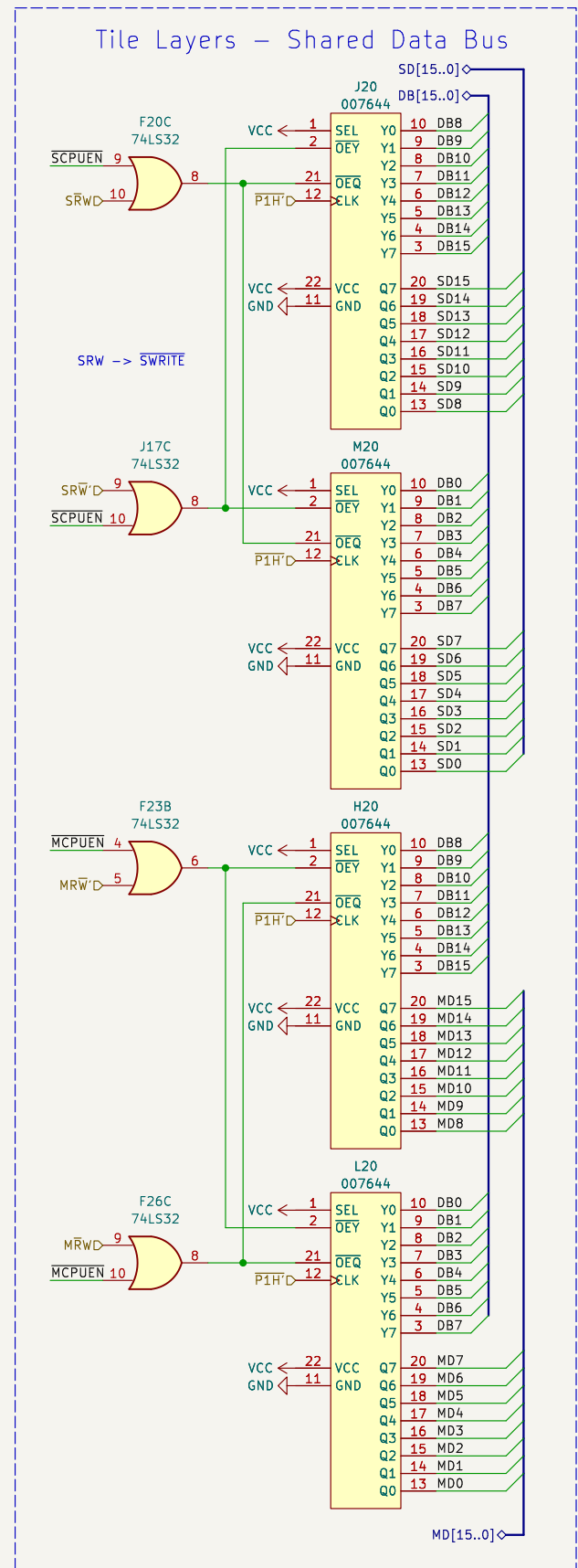
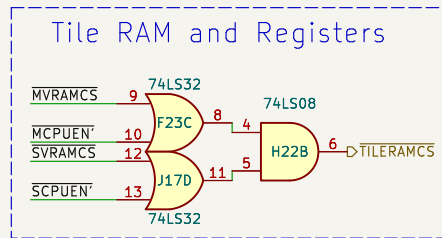
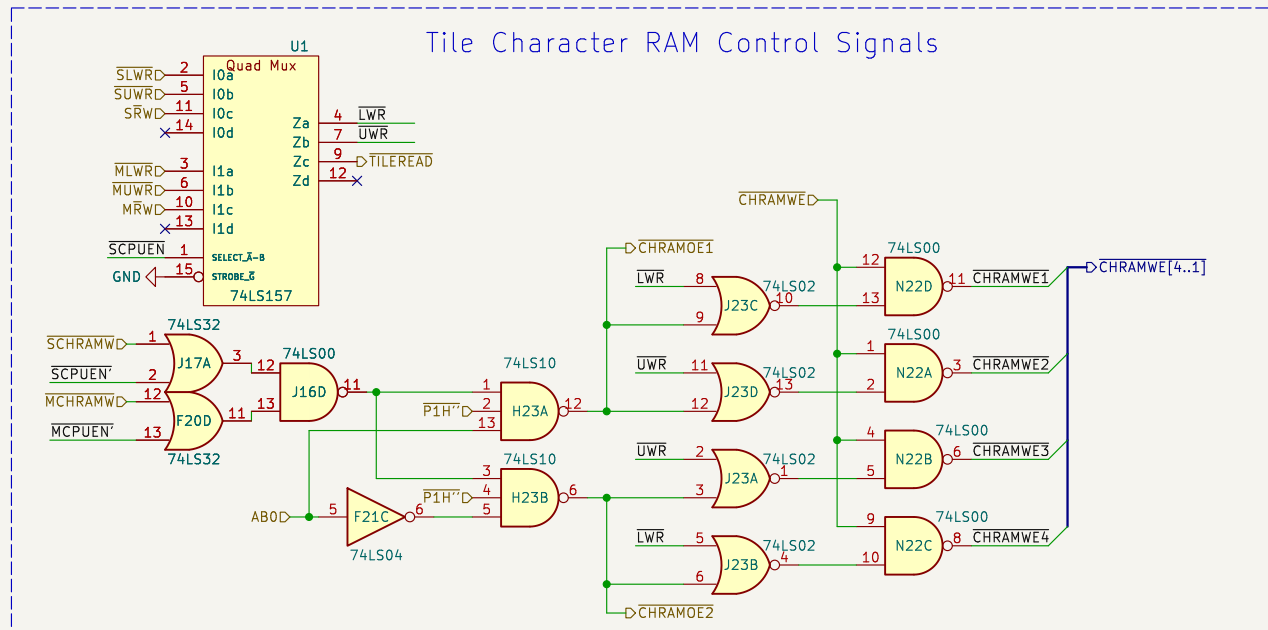
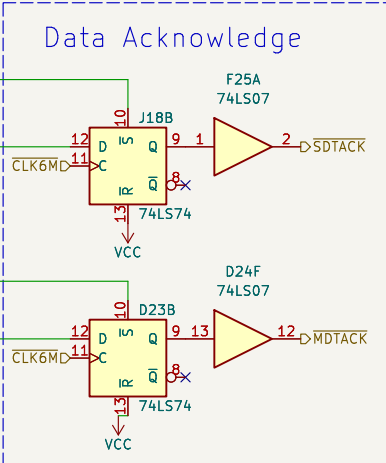
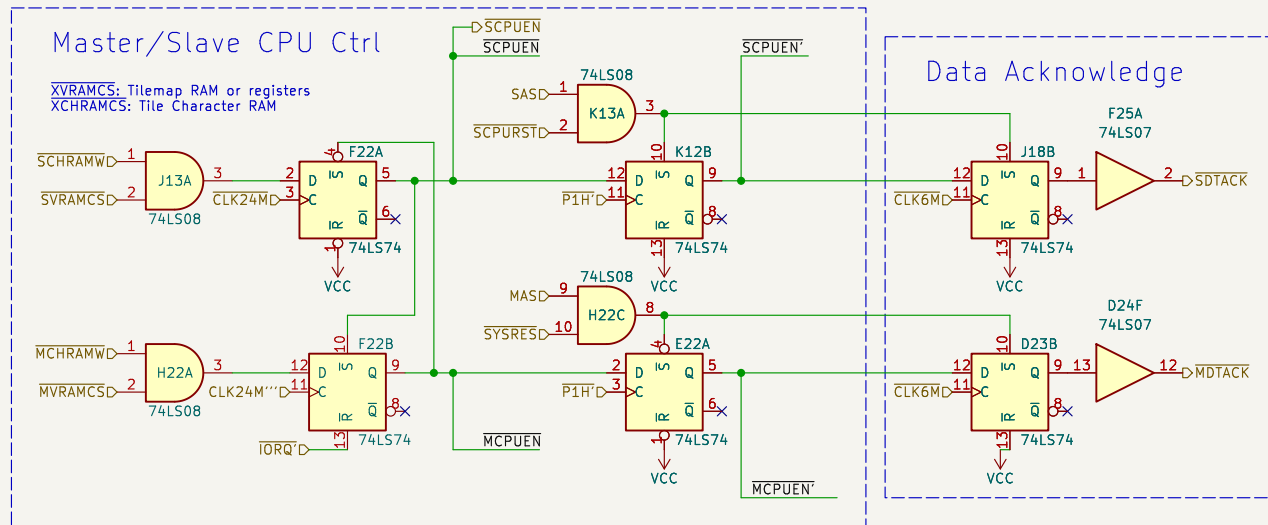










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Konami GX945

Sheet: /Tile Logic/

File: tile_logic.kicad_sch

Title: Gradius 3

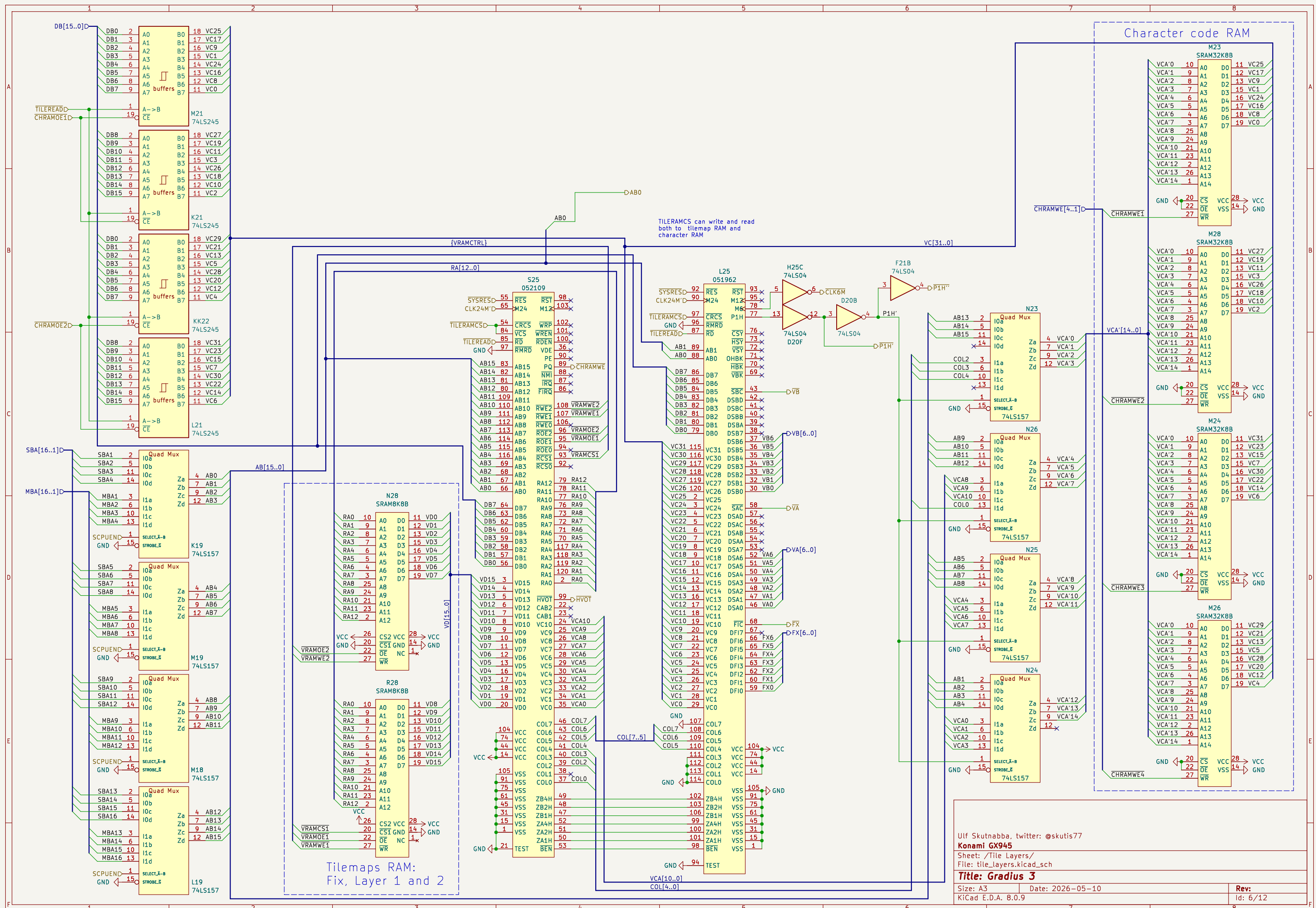
Size: A3

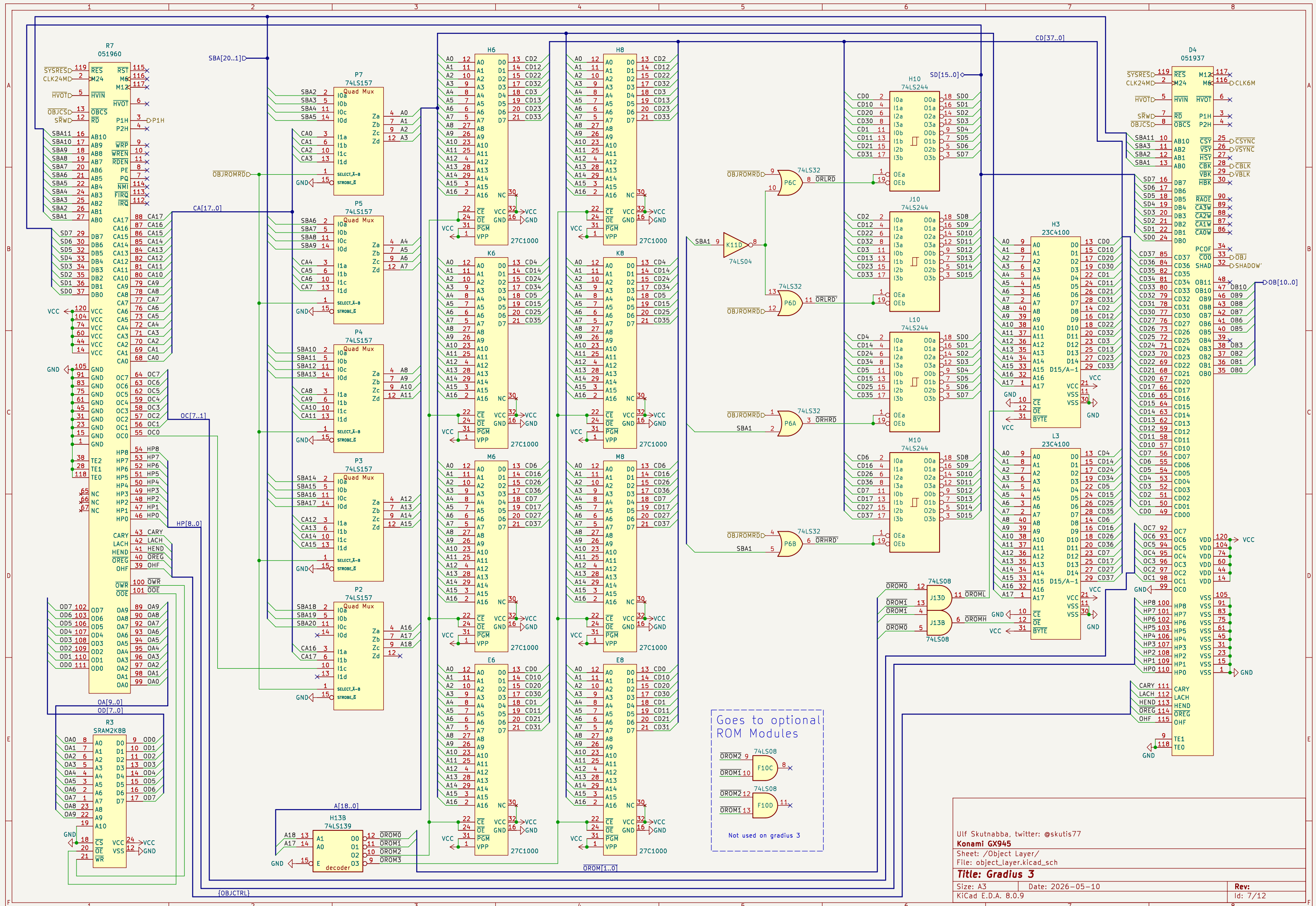
Date: 2026-05-10

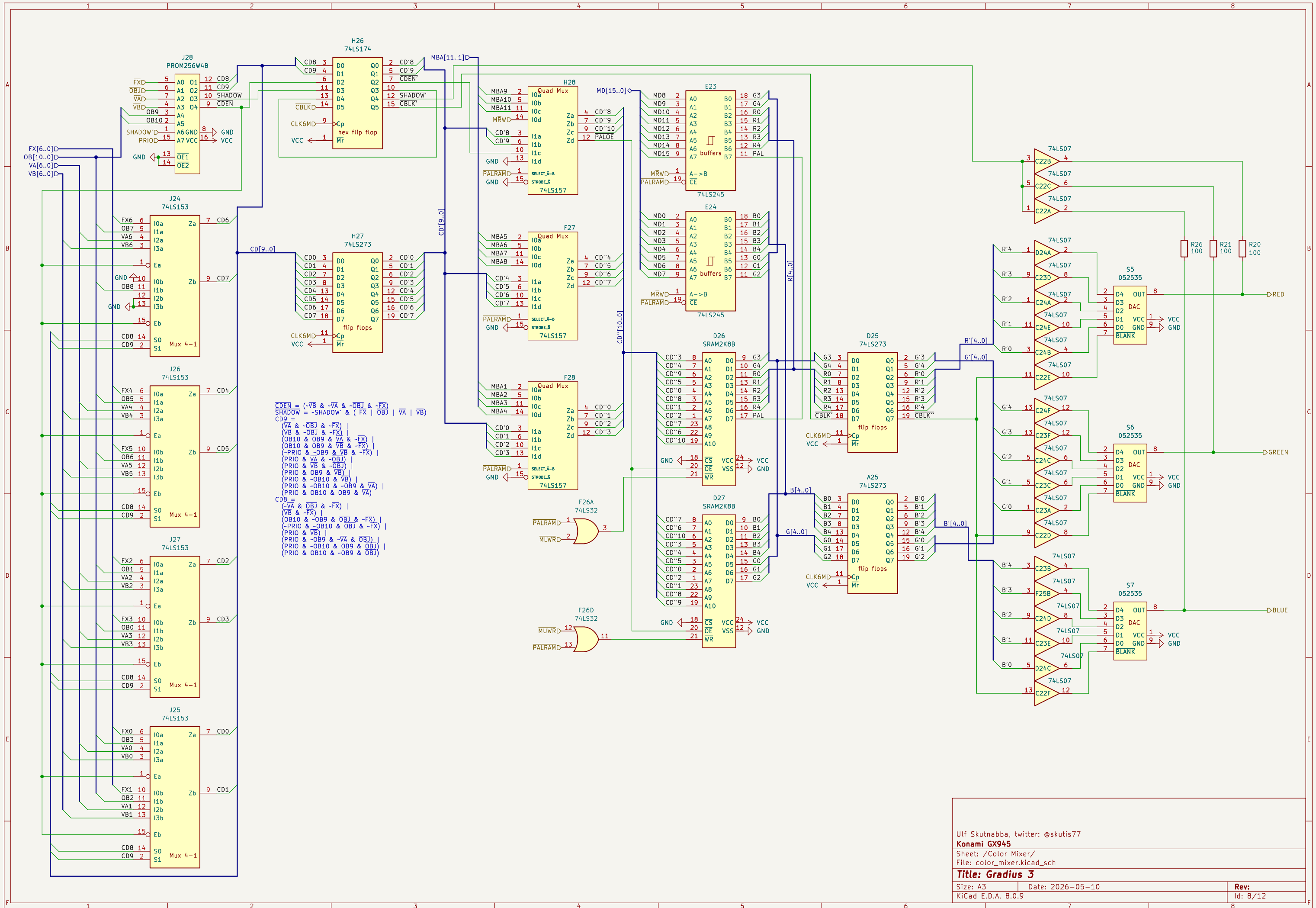
Rev:

KiCad E.D.A. 8.0.9

Id: 5/12







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Konami GX945

Sheet: /Color Mixer/

File: color_mixer.kicad_sch

Title: Gradius 3

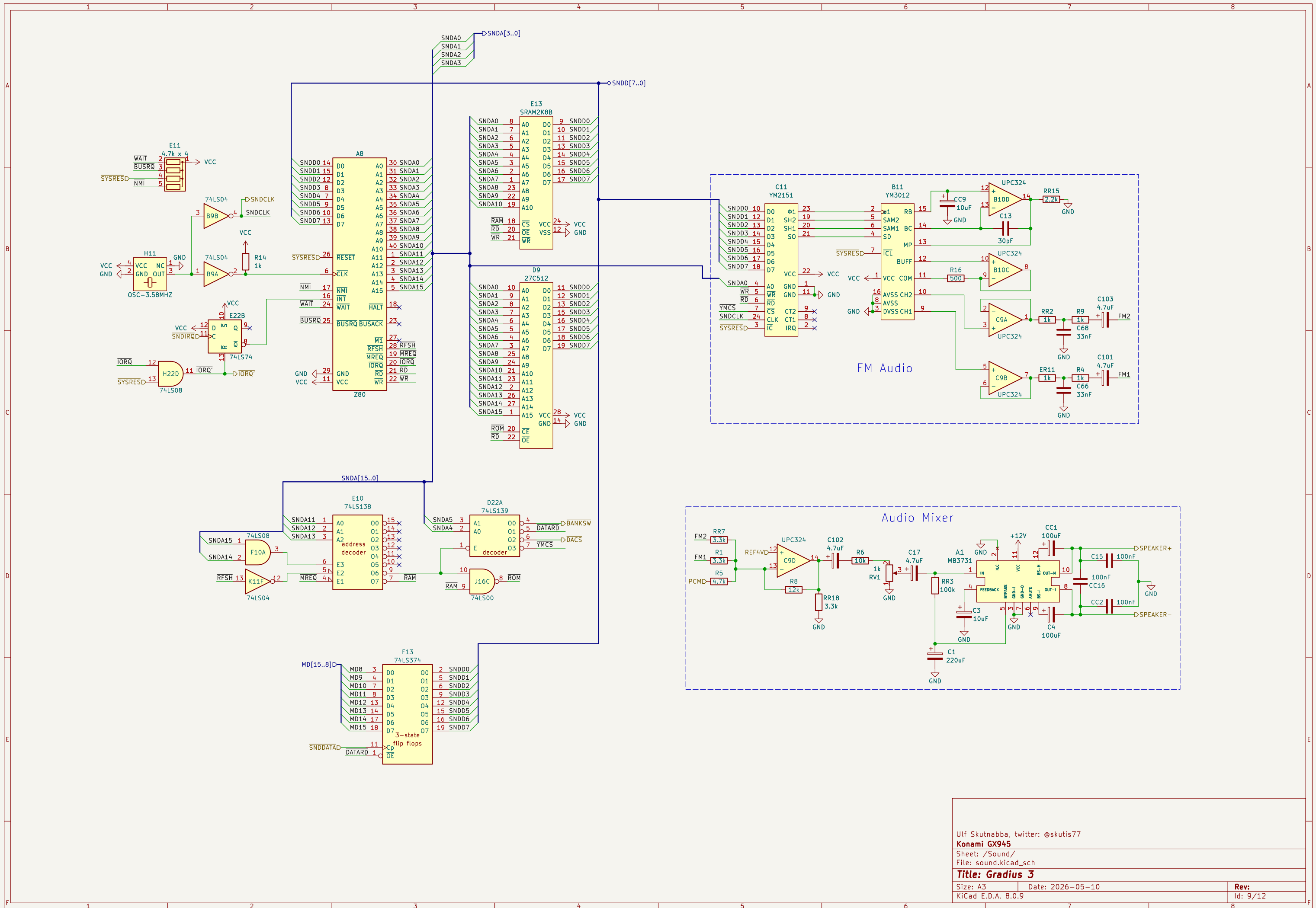
Size: A3

Date: 2026-05-10

Rev:

KiCad E.D.A. 8.0.9

Id: 8/12



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Konami GX945

Sheet: /Sound/

File: sound.kicad_sch

Title: Gradius 3

Size: A3 Date: 2026-05-10

KiCad E.D.A. 8.0.9

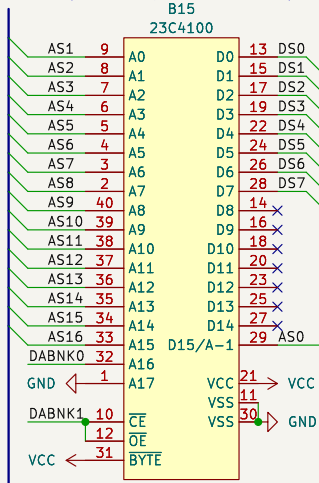
Rev:

Id: 9/12

B15 and B18 do not follow the standard Mask ROM behaviour. The BYTE mode select must be selected from factory and pin 31 is probably not connected.

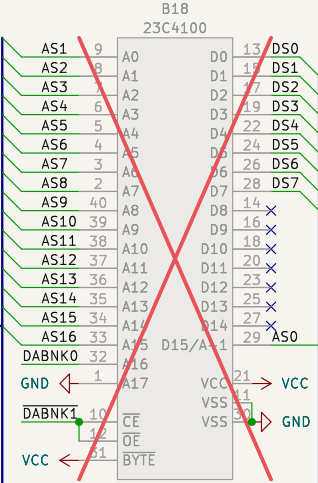
BANK A

Mounted on all releases except (Japan, program code S, split)

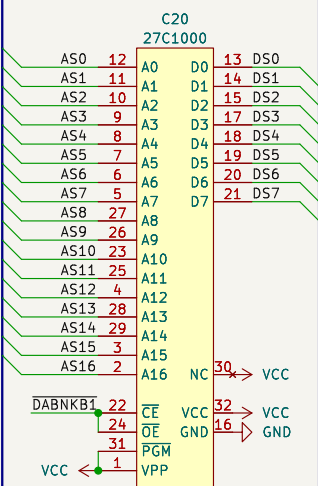
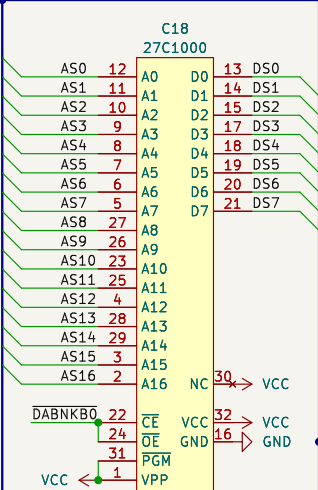
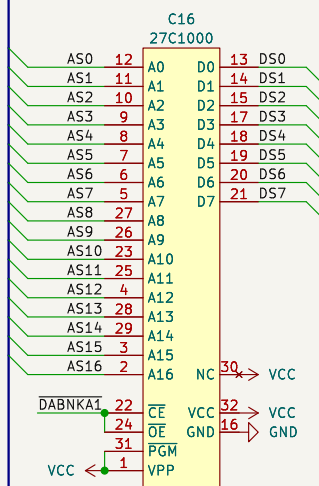
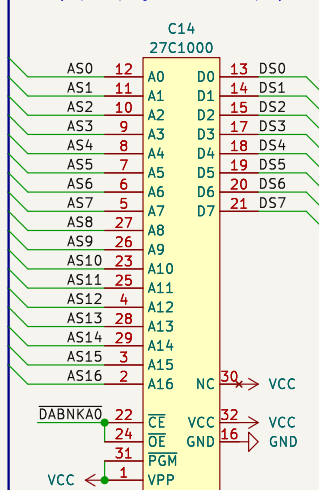


BANK B

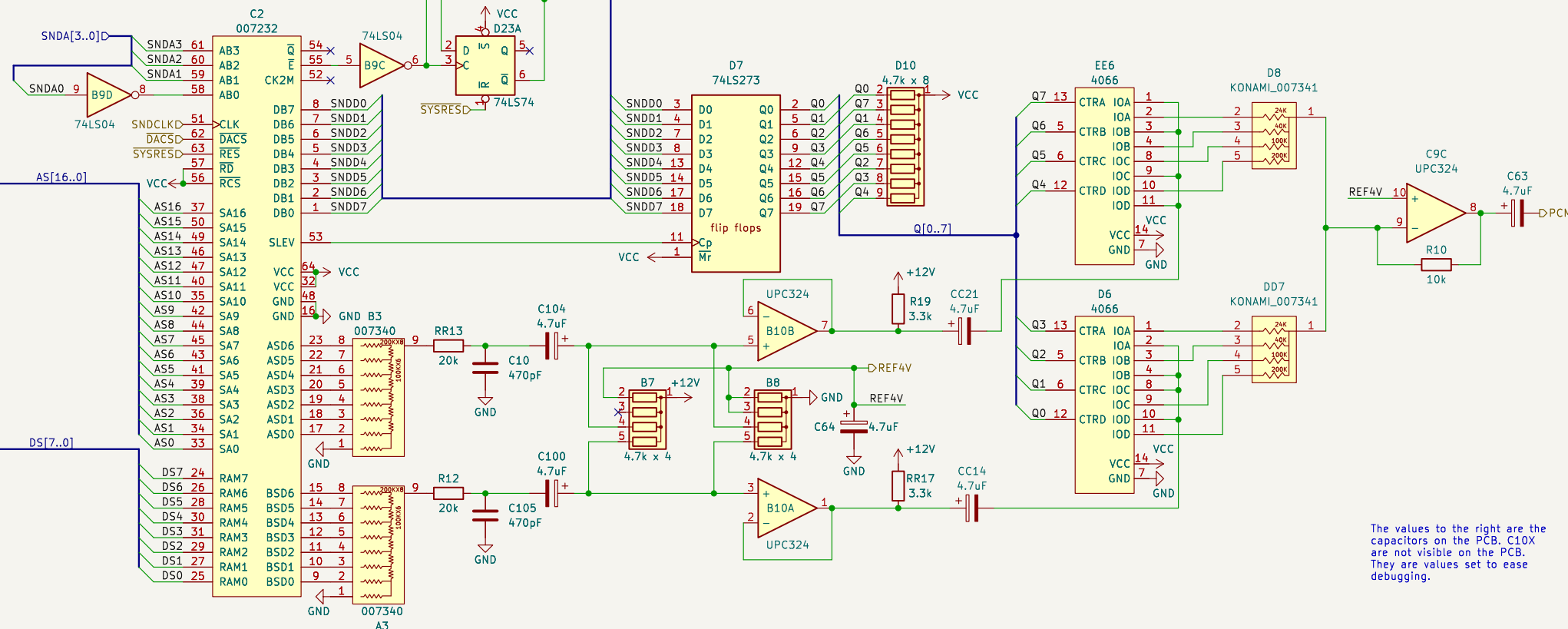
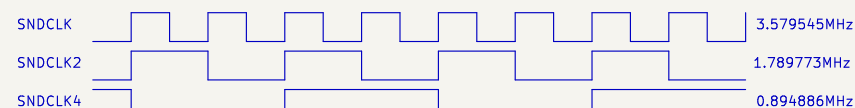
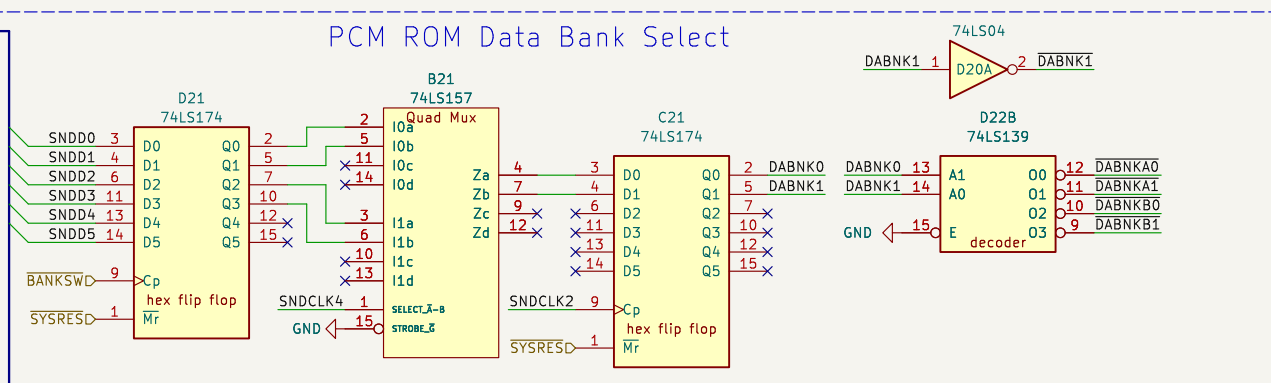
Not mounted on any releases



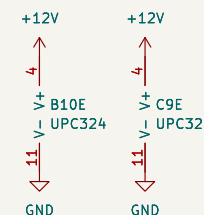
Mounted on Gradius III (Japan, program code S, split)



PCM ROM Data Bank Select



The values to the right are the capacitors on the PCB. C10X are not visible on the PCB. They are values set to ease debugging.



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Sheet: /PCM/

File: pcm.kicad_sch

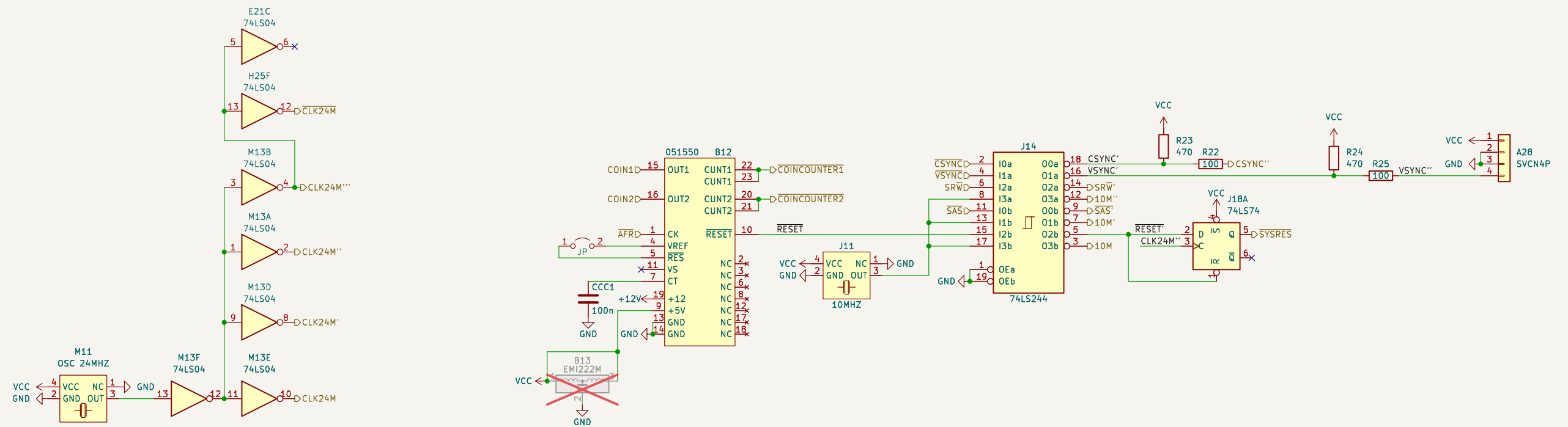
Title: Gradius 3

Size: A3 Date: 2026-05-10

KiCad E.D.A. 8.0.9

Rev:

Id: 10/12



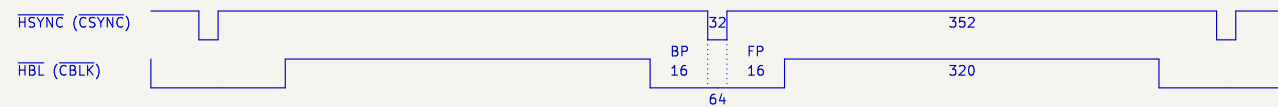
Horizontal and vertical synch timing diagrams

The pixel clock is derived from the 24MHz oscillator.
Pixel clock OVCK: $f = 24\text{MHz} / 4 = 6\text{MHz}$

The numbers in the HSYNC and HBL diagram are pixel clock cycles.
All edges are synchronised to the rising edge of the pixel clock.

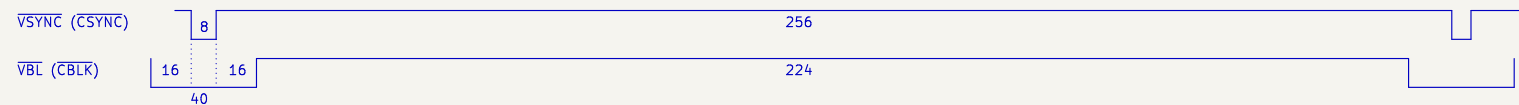
The signals have been measured at the output of the graphic chips.

If horizontal blanking is measured at the RGB DACs, the blanking is delayed 2 pixel clocks relative to composite sync. This gives BP = 14 and FP = 18.



HSYNC and HBL
Frequency $f = 6\text{MHz} / 384 = 15.625\text{kHz}$.
Period $T = 1/f = 64\mu\text{s}$.

The numbers in the VSYNC and VBL diagram are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.



VSYNC and VBL:
Frequency $f = 15.625\text{kHz} / 264 = 59.1856\text{Hz}$
Period $T = 1/f = 1 / 59.1856\text{Hz} = 16.896\text{ms}$

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Konami GX945

Sheet: /Misc/

File: misc.kicad_sch

Title: Gradius 3

Size: A3

Date: 2026-05-10

Rev:

KiCad E.D.A. 8.0.9

Id: 12/12